

KENAI 29 ENE, AK, USA

WMO: 703420

Lat: 60.7237N

Lon: 150.4484W

Elev: 282

StdP: 14.55

Time Zone: -9.00 (NAK)

Period: 10-19

WBAN: 26563

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-15.6	-27.3	1.2	-22.3	-20.4	1.8	-15.7	9.0	33.9	7.8	31.9	0.1	N/A	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	76.1	60.6	72.6	58.6	68.9	56.9	61.8	73.6	59.7	70.0	58.2	66.6	3.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.7	69.3	64.5	55.3	65.8	62.1	54.0	62.8	60.6	27.7	73.6	26.2	70.3	25.2	66.7	66.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	6.8	5.9	-24.1	80.1	7.9	4.4	-29.8	83.3	-34.4	85.8	-38.9	88.3	-44.6	91.5		
(5)			-24.4	63.1	7.8	2.0	-30.0	64.5	-34.6	65.6	-39.0	66.8	-44.7	68.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.5	15.6	19.8	23.4	35.6	45.2	52.8	56.9	54.8	47.0	36.2	20.3	17.7		
	DBStd	17.80	16.23	10.71	11.40	6.99	5.05	4.56	3.87	4.38	5.52	8.43	13.16	14.01		
	HDD50	5794	1065	846	825	432	161	19	1	8	116	430	890	1001		
	HDD65	10760	1530	1266	1290	882	613	368	253	317	541	894	1340	1466		
	CDD50	511	0	0	0	0	13	102	214	156	25	1	0	0		
	CDD65	4	0	0	0	0	0	1	2	1	0	0	0	0		
	CDH74	205	0	0	0	0	4	73	105	23	0	0	0	0		
	CDH80	30	0	0	0	0	0	10	20	0	0	0	0	0		
(14) Wind	WSAvg	2.0	1.4	2.0	2.1	2.4	2.8	2.4	2.1	2.2	2.1	1.9	1.6	1.5		
(15) Precipitation	PrecAvg	35.5	3.1	2.7	1.9	1.7	1.7	1.9	2.6	3.9	5.0	4.1	3.4	3.6		
	PrecMax	42.9	5.7	7.1	3.5	3.8	4.1	3.7	4.5	7.0	9.3	8.6	6.9	5.6		
	PrecMin	24.3	0.3	0.1	0.3	0.5	0.6	0.6	1.1	1.2	2.2	1.3	0.2	1.0		
	PrecStd	3.9	1.5	1.3	0.9	0.7	0.9	0.7	0.7	1.5	1.9	1.6	2.0	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.8	44.8	49.8	56.0	73.0	81.4	83.7	76.9	66.8	55.3	46.6	44.8		
		MCWB	39.6	39.0	41.2	44.1	54.5	61.4	65.4	62.1	54.4	47.6	40.1	38.8		
	2%	DB	42.3	40.9	46.0	52.8	68.0	75.3	76.6	73.0	62.8	52.2	43.3	41.5		
		MCWB	37.5	35.9	38.8	41.7	51.4	59.0	61.2	59.9	53.0	45.2	39.0	37.1		
	5%	DB	39.6	37.7	42.7	50.5	62.8	70.4	73.0	68.4	60.0	50.2	39.7	38.5		
		MCWB	35.7	34.2	36.9	40.5	49.0	56.1	60.1	57.3	51.9	44.0	37.4	35.2		
	10%	DB	36.8	34.6	39.6	48.6	58.9	66.5	69.5	65.6	57.2	48.1	36.9	34.9		
		MCWB	34.2	32.1	34.4	39.6	47.1	54.4	58.2	55.6	50.7	43.0	35.0	32.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.5	39.1	41.6	45.0	55.2	63.4	65.3	63.4	55.9	48.6	40.9	39.2		
		MCDB	45.0	43.7	47.9	54.0	72.6	79.4	82.4	75.1	63.0	52.0	44.5	43.0		
	2%	WB	37.7	36.4	39.2	42.6	52.4	59.6	62.5	60.5	54.3	46.9	39.3	37.1		
		MCDB	41.4	39.9	45.3	50.8	66.0	73.7	74.5	69.7	60.1	50.4	42.6	40.7		
	5%	WB	36.2	34.4	37.2	41.3	50.0	57.1	60.4	58.9	53.1	45.2	37.5	35.1		
		MCDB	39.5	37.5	42.1	49.0	60.7	68.4	71.1	66.8	58.0	48.7	39.7	38.1		
	10%	WB	33.9	32.3	35.0	40.1	48.1	54.9	58.9	57.2	51.9	43.7	35.1	32.7		
		MCDB	36.8	34.6	39.1	47.6	57.6	64.7	68.2	63.3	56.2	47.4	36.9	34.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	17.2	22.1	20.3	22.4	21.7	20.2	18.8	18.1	16.2	15.4	14.1		
		MCDBR	14.3	14.8	19.0	23.5	33.6	33.2	30.6	25.8	22.7	15.6	12.6	14.0		
		MCWBR	10.2	10.9	12.4	13.9	18.4	17.5	16.6	14.0	13.3	10.2	9.2	10.1		
	5% WB	MCDBR	13.8	14.0	18.0	20.6	28.7	28.7	27.1	21.5	17.8	13.4	11.9	13.3		
		MCWBR	9.8	10.6	12.2	12.4	16.4	16.0	15.2	12.5	11.5	9.3	8.8	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.235	0.283	0.299	0.341	0.368	0.357	0.369	0.353	0.320	0.296	0.246	0.220		
	taud		2.138	2.249	2.288	2.302	2.322	2.439	2.434	2.480	2.559	2.494	2.154	1.961		
	Ebn at Noon		167	226	263	271	271	277	270	266	258	225	156	108		
	Edh at Noon		12	21	28	34	36	32	32	28	22	16	12	9		
(44) All-Sky Solar Radiation	RadAvg		93	306	804	1305	1641	1713	1508	1197	777	369	134	47		
	RadStd		12	44	94	126	145	122	149	142	112	41	22	6		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (22 neighbors)	+1.24	+3.76	+3.26	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page